

The Factorion Key:

The Number 145 as a Harmonic Operator and Geometric Cipher in the Book of Taliesin

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Abstract

This paper proposes a novel hermeneutic framework for interpreting the medieval Welsh poems *Preiddeu Annwn* and *Cad Goddeu*, preserved in the Book of Taliesin. Building upon the geometric physics of Robert Edward Grant—specifically the Grant Polytope Framework and the Unity Harmonica—we introduce the number 145 (a “Factorion”) as a hidden Harmonic Operator applied to the text. While not explicitly written in the manuscript, 145 acts as a “grid-breaker” algorithm. We demonstrate that applying 145 to the poems’ structural constants unlocks a coherent narrative of dimensional extraction: the breaking of the 144-grid of the “Four-Cornered Fortress” ($145 \pmod{144} = 1$), the resolution of the Sothic time-loop ($1460 \approx 10 \times 145$), and the harmonic shattering of the “Glass Fortress.” Furthermore, we synthesize these findings with Somatic Bio-Geometric Harmonics, positioning the number 145 not merely as a mathematical abstraction but as an experiential resonance key—an “indexical cipher”—that bridges the gap between the material grid and the “Diamond Body” of human consciousness.

1 Introduction: The Intersection of Philology and Polymathic Physics

The study of the Book of Taliesin (Peniarth MS 2) has traditionally been the purview of philologists and Celtic historians, who view texts like *Preiddeu Annwn* (“The Spoils of Annwn”) and *Cad Goddeu* (“The Battle of the Trees”) as mythological fragments of a lost bardic tradition. However, a parallel tradition of interpretation exists—one that views these texts as repositories of a perennial philosophy encoded in the language of number and geometry.

This paper extends previous research identifying the “6,000 Sentinels” of *Preiddeu Annwn* as a scalar representation of the Fine-Structure Constant ($\alpha^{-1} \approx 137$) boundary condition, and the “Trees” of *Cad Goddeu* as frequency arrays in a geometric lattice. We specifically isolate the number 145 as a “Meta-Key.”

At first glance, the number 145 appears absent from the surface narrative. Yet, when analyzed through the multi-dimensional lens of Robert Edward Grant’s “Geometric Physics” and the “Somatic Bio-Geometric Harmonics” framework, 145 emerges as the “Harmonic Operator.” It functions as the mathematical crowbar used by the protagonist to pry the “Singularity” out of the “Grid.”

2 The Mathematics of the Factorion: A Code of Resurrection

To understand the application of 145, one must first define its unique number-theoretic properties within the context of Grant’s “Unity Harmonica.”

2.1 The Factorion Property

The number 145 is one of only four known numbers (along with 1, 2, and 40,585) equal to the sum of the factorials of its decimal digits.

$$1! + 4! + 5! = 1 + (4 \times 3 \times 2 \times 1) + (5 \times 4 \times 3 \times 2 \times 1) = 1 + 24 + 120 = 145$$

In the esoteric geometry favored by Grant, this property represents a “Perfect Return” or a “Resurrection Code.” It implies that a system can be decomposed into its constituent parts (digits/components), expanded through the dimension-building process of factorials, and return to its original unity without loss of information. This mirrors the shapeshifting magic of Taliesin in *Cad Goddeu*, wherein the bard is disassembled and reconfigured without loss of essence.

2.2 The Grid-Breaker Geometry (144 + 1)

In Grant’s Geometric Periodic Table, the number 144 (12×12) represents the “Wall of Light” or the perfect square of the material plane. It is the geometric limit of the “Four-Cornered Fortress.” Geometrically, 145 is the transcendence of this grid:

$$145 = 144 + 1$$

It represents the “Singularity” (+1) existing just outside the “Prison of the Cube” (144).

Crucially, this singularity is mirrored in the digital summation of the number itself. If we sum the digits ($1 + 4 + 5$), the result is 10, which harmonically reduces to 1 ($1 + 0 = 1$). This digital reduction mirrors the geometric extraction ($145 - 144 = 1$), reinforcing the role of 145 as a fractal expression of the “One” that transcends the system.

3 The Geometric Proof: The Pentagonal Key (P_{10})

While the Factorion property ($1! + 4! + 5!$) demonstrates the *arithmetic* self-reference of 145, the *geometric* authority of this number is derived from the Pentagram—the quintessential symbol of the “Christos” or Life Force (5).

To understand why 145 supersedes the 144 Grid, we must distinguish between two modes of growth: the **Metatronic (Fibonacci)** and the **Kristic (Krystal Spiral)**.

3.1 The Two Spirals: Entropy vs. Source

The 144 Grid is built on the Fibonacci sequence (1, 1, 2, 3, 5, 8...), which grows by accumulation—adding the past to the present to create the future. This creates a closed-loop system, often described in esoteric physics as a “Death Spiral” because it is detached from the center-point. 144 (F_{12}) represents the completion of this closed time-cycle (12×12).

In contrast, the **Kristic Sequence** (Krystal Spiral) grows through *doubling* (1, 2, 4, 8, 16...) and geometric expansion from the center, preserving a direct connection to Source. 145 belongs to the geometry of this “Open Source” reality.

3.2 145 as the 10th Pentagonal Number

The number 145 is the specific geometric lock that secures the Pentagon (5) into the Order of Completion (10).

- **The Pentagonal Formula:** Pentagonal numbers represent the accumulation of points required to build successively larger pentagons. The formula is:

$$P_n = \frac{n(3n-1)}{2}$$

- **The Calculation:** For the 10th step ($n = 10$):

$$P_{10} = \frac{10(30-1)}{2} = \frac{290}{2} = \mathbf{145}$$

Significance:

1. **The Number 5:** Represents the Pentagon/Flower/Life.
2. **The Number 10:** Represents the Decad/Divine Order.
3. **The Result (145):** This confirms that 145 is the “Perfected Pentagon.”

If 144 is the “Wall of Time” (The Square/Cube), 145 is the “Star” (The Pentagon) that opens the door. It sits exactly +1 step outside the 144 limit, confirming its role as the geometric vehicle for the “I AM” presence to exit the simulation grid.

3.3 The Hybrid Harmonic: Fibonacci and Lucas Braiding

While 144 is purely Fibonacci (F_{12}), 145 represents a “Hybrid Harmonic” that unifies the two primary growth sequences of nature: the **Fibonacci** (Life) and the **Lucas** (Structure) sequences.

$$145 = 5 \times 29$$

- **5** is the 5th Fibonacci Number (F_5).
- **29** is the 7th Lucas Number (L_7).

The Lucas sequence (2, 1, 3, 4, 7, 11, 18, 29...) defines the hidden structural framework of the Fibonacci growth. By fusing F_5 and L_7 , the number 145 mathematically “braids” the Life Force with the Structural Code, creating a unified geometry capable of holding both fluid and form.

4 The Cipher of the Four-Cornered Fortress

In *Preiddeu Annwn*, Arthur raids Caer Pedryvan, the “Four-Cornered Fortress.” Traditional analysis views this as a description of the world-square. However, applying the Factorion Key reveals a precise extraction algorithm.

4.1 The Modulo Operation

The fortress is “Four-Cornered” implies a cubic or square lattice structure, represented by the Grant Polytope number 144 (the squared geometry/Alphahedron edges). If we apply the Factorion (145) to the Fortress (144), we perform a modular operation:

$$145 \pmod{144} = 1$$

The “Remainder of One” is the Cauldron. The cipher reveals that the purpose of the raid is not conquest, but filtration: to extract the Singularity (1) from the Grid (144).

4.2 The Seven Survivors and the Fine Structure Limit

The poem repeatedly states, “Save seven, none returned.” This refrain anchors the mathematical extraction process:

“Three fullnesses of Prydwen we went into it;
Except seven none rose up from the Fortress of the Four Peaks.”
— *Preiddeu Annwn*, Stanza II

In physics, the Fine Structure Limit is often cited as 137. If we subtract this physical limit from our harmonic key:

$$145 - 137 = 8$$

The result, 8, corresponds to Arthur + 7 Survivors. The “Survivors” represent the margin of safety—the prime number stability required to navigate the gap between the Factorion Resurrection (145) and the Physical Limit (137).

5 The Dog-Head Frequency and the Time Loop

In *Cad Goddeu*, the narrator references combat with “Dog-Heads” (Cynbyn) on the “Mount of Eidyn.”

5.1 The Sirian Cycle

In esoteric astronomy, “Dog” invariably points to Sirius (The Dog Star). The Sothic Cycle of Sirius is exactly 1460 years. Applying the Factorion Key to the Sothic Cycle:

$$1460 \approx 10 \times 145$$

(Note: The product is exactly 1450, leaving a specific offset of 10 years, which aligns with the decimal harmonic scale). The “Dog-Heads” represent the time-cycle lock (1460). By defeating them, the enchanter Gwydion is breaking the time-loop. The Factorion (145) is the frequency that allows for a “decimal harmonic” escape from the cycle.

6 Somatic Integration: The Diamond Body (55) and the Forbidden Gap

This research integrates the Somatic Bio-Geometric Harmonics framework, which posits that these numbers are not merely literary devices but biological resonances. This synthesis of “experience” and “maths” is crucial

to understanding why 145 is the key, addressing the “Forbidden Gap” between biological life and the material grid.

6.1 The Osteological Cipher (55)

As established in Akasha Rose’s *The Harmonic of 55*, the human vessel is designed to resonate at 55 Hz.

- Spine (The Djed): 33 Vertebrae.
- Skull (The Temple): 22 Bones.
- Total Axis: $33 + 22 = 55$.

The number 55 is the 10th Fibonacci number (F_{10}). The number 144 is the 12th Fibonacci number (F_{12}).

The Lucas Connection (7:11): It is worth noting that the skull often encodes the Pi-ratio of 7:11 (as seen in the Great Pyramid). The numbers 7 and 11 are consecutive **Lucas Numbers** (L_4 and L_5). This suggests that while the body (55) is Fibonacci, the skeletal temple (skull) is constructed of Lucas numbers, reinforcing the “hybrid” nature of the vessel.

6.2 The Forbidden Gap: Bridging 5 and 8

The “Forbidden Gap” refers to a fundamental tension in crystallography and sacred geometry between the Pentagonal Life Force (5) and the Octagonal/Cubic Grid (8).

- **The Problem of 5 (Life):** In classical crystallography, 5-fold symmetry is “forbidden” because pentagons cannot tile a flat plane (or fill 3D space) without leaving gaps. Biological life, however, relies on 5-fold symmetry (DNA cross-section, hands, flowers).
- **The Problem of 8 (Grid):** The inorganic world, specifically the crystal lattice, relies on 8-fold or 4-fold symmetries (The Cube/Octahedron) which act as the stable “prison” or grid of matter.
- **The Gap:** There is a geometric dissonance between the fluid, pentagonal nature of life (5) and the rigid, octagonal nature of the material grid (8). Life (5) cannot naturally “dock” into the Grid (8).

The Solution is 55: The number 55 acts as the harmonic bridge that resolves this gap.

- **Fibonacci Sequence:** 55 is the 10th Fibonacci number (F_{10}). The Fibonacci sequence (1, 1, 2, 3, 5, 8, 13, 21, 34, 55...) is the mathematical progression that allows integers to approximate the Golden Ratio (ϕ), which drives 5-fold symmetry.
- **Harmonic Product:** 55 is the product of the Life Force (5) and the Master Number of Structure (11).

$$5 \times 11 = 55$$

By resonating at 55 (The Diamond Body), the human vessel effectively “tunes” the Pentagonal Life Force (5) to a frequency that can harmonically interact with the Grid, allowing the “Singularity” (145) to enter the system. The “Forbidden Gap” is bridged not by force, but by the specific harmonic of 55.

6.3 The Solar Calculation (111)

Cad Goddeu does not enumerate a fixed or canonical list of trees. Within the present framework, the "Tree List" is treated not as a historical catalogue but as a structural-symbolic array generated by the **Solar Prime 29**. As established in the "Archeocryptographic Revaluation" [?], a strict philological audit identifies 30 explicit plant terms, which reduce to the operational prime 29 when the genus *Ulex* (Furze/Gorse) is unified. This prime 29 is the generator of the Factorion Key 145 via the 10th Pentagonal Number (P_{10}).

The stabilizing Solar Core frequency (111) is derived not from a subtraction of trees, but from the activation of the Diamond Body (55) itself:

$$(55 \times 2) + 1 = 111$$

Here, 55 represents the somatic vessel (F_{10}), the doubling represents the twin polarity of the electric and magnetic fields, and the +1 represents the extracted Singularity. This calculation confirms that 111 is the fundamental harmonic constant required to stabilize the high-voltage 145 state within the biological form.

Validation: The Magic Square of the Sun This "Solar" designation is rigorously confirmed by the arithmetic of the 6×6 **Magic Square of the Sun**. In this classical grid, which contains the integers 1 to 36 and sums to 666 (The Solar Number), the sum of any single row or column is exactly **111**. This proves that 111 is not an arbitrary result, but the Fundamental Harmonic Constant of the Solar Matrix.

The Interplay of Fifths (The "Year and a Half" Code): In *Cad Goddeu*, Taliesin says "I have been the light of lanterns / A year and a half." Numerically, 1.5 corresponds to the ratio 3:2 (The Perfect Fifth). Taliesin is the dynamic force (the Spiral).

Why 111 is the Final Key: While 1.5 is the motion, 111 is the resolution.

- The Diamond Body is 55.
- The Twin Polarity (Matter/Antimatter) is $55 \times 2 = 110$.
- The Singularity Spark (+1) makes $110 + 1 = 111$.

7 Conclusion: The 145 Protocol

This paper proposes that the number 145 serves as a "Meta-Key" for the Book of Taliesin. It is not merely a number but a protocol for Dimensional Escape:

1. **Identify the Grid (Mod 144):** Recognize the material prison of Caer Pedryvan. This is the Alphahedron or Carbon lattice of the physical world.
2. **Bridge the Gap (55):** Activate the Somatic Harmonic of 55 Hz (The Diamond Body) to resolve the dissonance between Life (5) and Grid (8).
3. **Introduce the Singularity (+1):** Apply the Factorion (145) to extract the Consciousness/Cauldron from the Grid.
4. **Resonate the Wall (/60):** Shatter the Sentinel boundary (6,000) via harmonic dissonance.

The ratio $145/60 \approx 2.416$ corresponds to a Silver Ratio convergent, creating a specific tension that destabilizes the "Glass" boundary of the 60-base structure (137 Limit).

The final result of this protocol is the liberation of the Diamond Body Singularity: a state of consciousness that has integrated the biological life force (5) with the structural grid (8) through the harmonic of 55, and then transcended the grid entirely ($144 + 1$) to operate as a free agent within the multidimensional field.

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